

Esperanza condicionada a una σ -álgebra independiente

Proposición 1. Sea $X \in \mathcal{L}^1(\mathbb{P})$ y $\mathcal{F}$ una σ -álgebra independiente de $\sigma(X)$

$$\implies \mathbb{E}[X \mid \mathcal{F}] \equiv \mathbb{E}[X] \text{ c.s.}$$

Demostración: Sea $Z \in \mathcal{L}^\infty(\mathbb{P})$ $\mathcal{F}$ -medible, entonces $\mathbb{E}[Z \cdot \mathbb{E}[X \mid \mathcal{F}]] = \mathbb{E}[X \cdot Z]$. Como $\sigma(X)$ es independiente de $\mathcal{F}$, Z y X son independientes, por lo que

$$\mathbb{E}[Z \cdot \mathbb{E}[X \mid \mathcal{F}]] = \mathbb{E}[Z \cdot X] \stackrel{\text{prop-esperanza-prod-var-aleatorias-indep/??}}{\downarrow} \mathbb{E}[Z] \cdot \mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X] \cdot Z] \stackrel{\text{lineal}}{\uparrow}$$

Entonces, la función $Y := \mathbb{E}[X] \mathbb{1}_\Omega$ es $\mathcal{F}$ -medible y cumple $\mathbb{E}[Z \cdot Y] = \mathbb{E}[X \cdot Z]$ para todo Z $\mathcal{F}$ -medible. Por la unicidad de la esperanza condicionada, $Y = \mathbb{E}[X \mid \mathcal{F}]$ c.s. ■

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